

RICARDO JOSÉ SUÁREZ RAVELO

Valledupar, Colombia · www.linkedin.com/in/ricarsur/ · +57 3006282873 · ricarsurbusiness@gmail.com · https://ricarsurbusiness.github.io/my_portfolio

Full Stack Developer with a strong backend focus, specialized in building functional and efficient solutions. Hands-on experience with Go, Node.js, Python, React, and TypeScript, working on both API development and user interfaces. Familiar with different database systems (PostgreSQL, MongoDB, MySQL) and modern development tools such as Docker for deployment. Seeking to apply my technical skills in challenging projects where I can continue growing professionally.

PROFESSIONAL EXPERIENCE

SIMIL Solution SAS - Qflow

Backend Developer - Intern

**Valledupar, Cesar,
Colombia**

August 2025– present

- Led the backend development of QFlow, an enterprise queue management system, implementing a microservices architecture with Go that processes 1,000+ daily transactions with 95% functional completeness.
- Designed and implemented the critical queue module with executive assignment, creating 24 repository methods and 13 RESTful endpoints, achieving thread-safe concurrency with optimistic locking in PostgreSQL.
- Developed a multi-channel notification system integrating Twilio API, implementing an automatic fallback strategy that reduced messaging costs by 50% and improved deliverability to 99%.
- Optimized database performance by designing 13 partial indexes and specialized queries, reducing queue query times from >1s to <150ms at the 95th percentile.
- Implemented real-time communication using WebSocket/SSE for 100+ concurrent users, integrating an asynchronous worker system with a pool of 5 goroutines for parallel processing.
- Set up a complete deployment pipeline using Docker, KrakenD API Gateway, and managing 5 independent microservices with their respective PostgreSQL databases.

SIMIL Solution SAS - Mussico

Backend Developer

**Valledupar, Cesar,
Colombia**

February 2024 – June 2024

- Developed a centralized General API for the Mussico platform using Node.js with Express.js, implementing a RESTful architecture that unified multiple functionalities into a single access point.
- Designed and implemented a JWT authentication system with role-based authorization middleware, ensuring secure access to protected resources and token validation on every critical request.

- Integrated Azure Blob Storage for media file management, developing a scalable storage service that handles upload, download, and management of profile images and music content using temporary signed URLs.
- Architected a NoSQL database with MongoDB, designing optimized schemas with references and embedded documents, and implementing compound indexes to improve search and filtering query performance.
- Implemented a layered architecture following SOLID principles, separating routes, controllers, services, and repositories, with data validation via middleware and centralized async error handling.
- Configured a containerization pipeline with Docker and Docker Compose, orchestrating multiple services (API, MongoDB, Redis) with secure environment variables and persistent volumes for development and production environments.

Personal Project
Full Stack developer

Colombia
July 2025

- Designed a scalable microservices architecture with FastAPI (Python), implementing 4 independent services (Auth, Products, Cart, Orders) with an API Gateway, handling multiple concurrent requests.
- Built a robust backend with JWT authentication, data validation using Pydantic, SQLAlchemy ORM, and a MySQL 8.0 database with optimized schemas and unit tests (45+ test cases with 100% coverage).
- Developed a modern frontend with React 19 + Vite, reusable components (Login, Register, Shop, Cart, ProductList), API integration, and session persistence in localStorage.
- Implemented inter-service communication via internal HTTP between containers, allowing the Cart Service to query the Auth Service and the Order Service to interact with multiple services.
- Full orchestration with Docker and Docker Compose, configuring 6 interconnected containers, persistent volumes, CORS, environment variables, and interactive documentation at /docs.
- Features: Secure registration/login, product catalog, shopping cart with automatic calculations, order processing, responsive interface with CSS3.

Technologies: FastAPI, Python, React, Vite, MySQL, JWT, Docker, Pytest, SQLAlchemy

EDUCATION		
UNIVERSIDAD POPULAR DEL CESAR		Valledupar, Cesar,
Systems Engineering		Colombia
		2019-2025
GOOGLE CLOUD		Virtual
Build a Secure Google Cloud Network Skill Badge		2024

CISCO
CCNA: Introduction to Networks

Virtual
2023

CISCO
Python Essentials 1

Virtual
2023

SKILLS

Technical: Node.js, TypeScript, JavaScript, Go, Python, Express, Docker, MongoDB, PostgreSQL, MySQL, React, Flutter, Dart, Astro, Jira

Soft Skills: Problem solving, critical thinking, effective communication, teamwork, adaptability, time management, proactivity, attention to detail, analytical thinking, collaboration in agile environments (Scrum/Kanban)

LANGUAGES

- Spanish: Native
- Portuguese: Advanced
- English: Intermediate